

The measured contact current of relay K1 is shown in Fig. 3. From the figure, it can be found that relay K1 has its peak contact current within a period of time after conduction. Since the maximum current the relay can withstand is 10A (according to the specification), and the surge current is way more than 10A, it may damage the relay contacts and lead to a short-circuit.

The cause of surge current

Troubleshooting revealed that the reason for the relay contact short-circuit was that the surge current flowing through the relay was too large. In the circuit in Fig. 2, what was the reason for the surge current during relay attraction? There are three possibilities.

1. If the DM inductor of the π filter is saturated, it will lose the suppression of current
2. If the capacitance of filter capacitor C1 is large, the distortion in the current pulse will also be large
3. If the current sampling resistance of PFC is small, the input current reaches the clamp point when the PFC starts working

Apart from the surge current, there are other reasons why a relay might fail, and it is important to address them too. Some of these reasons include redundant objects inside the relay, dirt on the contact material, improper process structure, improper welding, contact ablation, adhesion, silver ion migration, and reed displacement caused by external application.

Most of these failures are caused by improper control of the relay production process. Therefore, it is critical for relay manufacturers to improve the production environment, quality control, and inspection systems to prevent frequent relay failures.

How to overcome these problems?

Designers need to be extremely careful while designing PFC circuits



Fig. 4: A bad contact weld (Credit: <https://www.pickeringtest.com>)



Fig. 5: PT570730 relay with silver contacts (Credit: element14)

and must make sure to optimise the parameters given above. For example, selecting the DM inductor with a larger saturation current and reducing the capacitance of the first capacitor in the π filter would be beneficial. Increasing the PFC current sampling resistance and reducing the PFC clamping current would also help.

In addition to these, the selection of the relay is also critical. Relay contacts are made of different material, each suited for different application. Designers must first choose the relay according to the actual use requirements, and then carefully determine the required functional



Fig. 6: Finder 60 series general-purpose relay with AgNi contacts (Credit: Finder)



Fig. 7: HF46F-G/012-HS1T Hongfa relay with AgSnO₂ contacts (Credit: Hongfa)

and physical characteristics while selecting the relay to avoid failure during use.

Relay contact materials

Here are some different relay contact materials and their properties.

Fine silver. Among all metals, fine silver has the best electrical and thermal characteristics and so is the best all-around material in the market. When used in relays, it